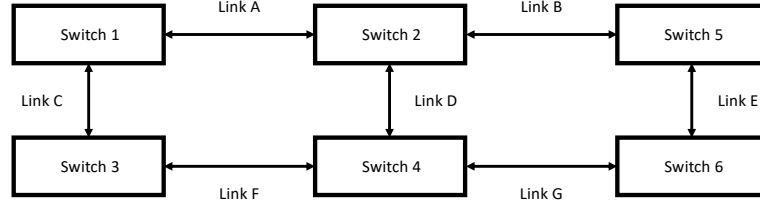


COMPUTER SCIENCE TRIPOS Part IB – 2020 – Paper 5

4 Computer Networking (awm22)



The network illustrated above represents an Ethernet Layer-2 network that uses spanning-tree to compute forwarding tables. Assume all links have a link-weight of one.

[Note: Tie-breaking/leader-elections use the switch identifier from this diagram.]

Topic 3 data-link spanning tree protocol and switch architecture (forwarding tables.)

(a) Compute the steady state routing/forwarding table for Switch 3.

Answer:

N.B. This set of answers was updated from those originally posted with the exam paper. It reconciles an error in the solution of part (a) that also impacted the published solution to (b). These two answers have been replaced. Apologies for the confusion, awm22.
(my thanks to Ethan G. Jones)

1. Running spanning-tree:

Notation: (root, distance, root via switch #)

Initially every switch proposes itself as root	S1: (1, 0, -) ↔ S2: (2, 0, -) ↔ S5: (5, 0, -)
	↓ ↓ ↓
	S3: (3, 0, -) ↔ S4: (4, 0, -) ↔ S6: (6, 0, -)
Switches 2 and 3 accept Switch 1 as root	S1: (1, 0, -) ↔ S2: (1, 1, 1) ↔ S5: (2, 1, 2)
	↓ ↓ ↓
	S3: (1, 1, 1) ↔ S4: (2, 1, 2) ↔ S6: (4, 1, 4)
Switches 4 and 5 accept Switch 1 as root	S1: (1, 0, -) ↔ S2: (1, 1, 1) ↔ S5: (1, 2, 2)
	↓ ↓ ↓
	S3: (1, 1, 1) ↔ S4: (1, 2, 2) ↔ S6: (2, 2, 4)
Finally Switch 6 accepts Switch 1 as root via Switch 4	S1: (1, 0, -) ↔ S2: (1, 1, 1) ↔ S5: (1, 2, 2)
	↓ ↓ ↓
	S3: (1, 1, 1) ↔ S4: (1, 2, 2) ↔ S6: (1, 3, 4)
Final result	Switch 1 ↔ Switch 2 ↔ Switch 5
	↓ ↓ ↓
	Switch 3 ↔ Switch 4 ↔ Switch 6

Therefore Switch 3 will send all its traffic destined for another switch via Link C to Switch 1 which becomes the root. Link F connecting it to Switch 4 will be disabled as Switch 3 can directly connect to the root and Switch 4 will prefer to connect via Switch 2.

Switch 3's forwarding table:

Destination	Link
Switch 1	Link C (→ S1)
Switch 2	Link C (→ S1)
Switch 3	-
Switch 4	Link C (→ S1)
Switch 5	Link C (→ S1)
Switch 6	Link C (→ S1)

[4 marks]

- (b) Noting which switches recompute a solution, enumerate the changed forwarding tables in switches of this network resulting from the complete failure and removal of link D.

Answer:

2. Removing Link D (Switch 2 to Switch 4):

Switch 4 will propose itself as root after noticing Link D is gone	S1: (1, 0, -) ↓ S3: (1, 1, 1)	↔	S2: (1, 1, 1) ↓ S4: (4, 0, 4)	↔	S5: (1, 2, 2) ↓ S6: (1, 3, 4)
Switch 6 will accept 4 as the root and pass this onto Switch 5. Switch 3 will reply to 4 with it's path to Switch 1.	S1: (1, 0, -) ↓ S3: (1, 1, 1)	↔	S2: (1, 1, 1) ↓ S4: (4, 0, 4)	↔	S5: (1, 2, 2) ↓ S6: (4, 1, 4)
Switch 4 will accept 1 as the root via 3 and forward to 6. Switch 5 will reply to 6 with it's root to Switch 1.	S1: (1, 0, -) ↓ S3: (1, 1, 1)	↔	S2: (1, 1, 1) ↓ S4: (1, 2, 3)	↔	S5: (1, 2, 2) ↓ S6: (4, 1, 4)
Switch 6 will accept Switch 1 as root and chose to forward via 4	S1: (1, 0, -) ↓ S3: (1, 1, 1)	↔	S2: (1, 1, 1) ↓ S4: (1, 2, 3)	↔	S5: (1, 2, 2) ↓ S6: (1, 3, 4)
Final result	Switch 1 ↓ Switch 3	↔	Switch 2 ↓ Switch 4	↔	Switch 5 ↓ Switch 6

Therefore Switch 4 will now send its traffic not for itself or Switch 6 to Switch 3 instead of Switch 2 and Link E (Switch 5 to 6) will stay disabled. Switch 2 will now forward traffic for Switch 4 or 6 to Switch 1 (instead of Switch 4), Switch 1 will pass it to Switch 3 (instead of Switch 2) and Switch 3 will pass it to Switch 4 (instead of Switch 1).

Switch 1's new forwarding table:

Destination	Link
Switch 1	-
Switch 2	Link A (→ S2)
Switch 3	Link C (→ S3)
Switch 4	Link C (→ S3)
Switch 5	Link A (→ S2)
Switch 6	Link C (→ S3)

Switch 2's new forwarding table:

Destination	Link
Switch 1	Link A (→ S1)
Switch 2	-
Switch 3	Link A (→ S1)
Switch 4	Link A (→ S1)
Switch 5	Link B (→ S5)
Switch 6	Link A (→ S1)

Switch 3's new forwarding table:

Destination	Link
Switch 1	Link C (→ S1)
Switch 2	Link C (→ S1)
Switch 3	-
Switch 4	Link F (→ S4)
Switch 5	Link C (→ S1)
Switch 6	Link F (→ S4)

Switch 4's new forwarding table:

Destination	Link
Switch 1	Link F (→ S3)
Switch 2	Link F (→ S3)
Switch 3	Link F (→ S3)
Switch 4	-
Switch 5	Link F (→ S3)
Switch 6	Link G (→ S6)

[9 marks]

- (c) Following the removal of Link D, a new link H is added between Switch 1 and 4; however, this link fails frequently.

Denied access to monitor the network-traffic, outline a diagnostic strategy to identify the faulty Link H, making clear how the network-operator might use interrogation of network switch forwarding tables.

Answer:

no Wireshark

Simply regularly looking at the tables of switches is enough [1 mark]; while some switches are not affected [1 mark], the impacts will be clear by virtue of seeing adjacencies (switch 1 and switch 4) appear and disappear)

(recording changes in the full graph is enough but may not practically be available) [bonus 1 where appropriate]

Other additional marks for each of

aliasing (not seeing the tables sufficiently often)

or mention of route-flap damping as a related (but not the same) issue... causing such links to be disabled outright.

[4 marks]

- (d) Now suppose the switches do not permit interrogation of switch forwarding tables and that the link status information is untrustworthy. Outline other techniques that might be used to identify the failed link.

[3 marks]

Answer: while some insights are circumstantial:

(CPU usage climbing because the need to recompute answers)

answers are expected to be “look at the traffic on the links” or “look at the spanning-tree communications” to reconstruct what is happening.

Interesting additional suggestions included using hosts to create arbitrary evaluation matrices of test traffic, incorporating a purposefully low-capacity link, interrogating outcomes and comparing with physical manipulation of the links... (other approaches may be appropriate.)
